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Fractional order heat conduction law in magneto-thermoelasticity involving two temperatures

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Abstract. A new mathematical model of two-temperature magneto-thermoelasticity is constructed where the fractional order heat conduction law is considered. The state space approach is adopted for the solution of one-dimensional application for a perfect conducting half-space of elastic material with heat sources distribution in the presence of a transverse magnetic field. The Laplace-transform technique is used. A numerical method is employed for the inversion of the Laplace transforms. According to the numerical results and its graphs, conclusions about the new theory are given. Some comparisons are shown in figures to estimate the effects of the temperature discrepancy and the fractional order parameter on all the studied fields.

Mathematics Subject Classification (2000). 74Bxx · 35Qxx · 65Nxx · 65Txx.

Keywords. Magneto-thermoelasticity · Two-temperature theory · Fourier's law · Modified Riemann-Liouville fractional derivative · New fractional Taylor's series · State space approach · Laplace transforms · Fractional calculus.

List of symbols

a	temperature discrepancy
B_i	components of magnetic field strength
c	the speed of light
c_o	$\sqrt{\frac{\lambda+2\mu}{\rho}}$, speed of propagation of isothermal elastic waves
C_E	specific heat at constant strain
e	dilatation
e_{ij}	components of strain tensor
E_i	components of electric field vector
$H(.)$	Heaviside unit step function
H_i	magnetic field intensity
J_i	components electric density vector
k	thermal conductivity
$\boldsymbol{q}$	heat flux vector
Q	the intensity of applied heat source per unit volume
t	time
T	absolute thermodynamic temperature
$T_\circ$	φ_0 reference temperature
u_i	components of displacement vector
α, υ	fractional orders
α_0	$\sqrt{\mu_0 H_0^2/\rho}$, Alfven velocity
α_T	coefficient of linear thermal expansion
α^*	$1+(\alpha_0/c)^2$
β	$(\alpha_0/c_0)^2$
β_0	dimensionless temperature discrepancy

β_0^c	the critical value of β_0
γ	$(3\lambda + 2\mu)\alpha_T$
$\delta(.)$	Dirac delta function
δ_{ij}	Kronecker's delta
ε_{ij}	components of strain tensor
ε_0	electric permittivity
ε	thermal coupling parameter
η_0	$\frac{\rho C_E}{k}$
θ	$T - \varphi_0$; $\left\|\frac{\theta}{\varphi_0}\right\| << 1$
$\lambda,\ \mu$	Lame' constants
μ_0	magnetic permeability
ρ	mass Density
σ_{ij}	components of stress tensor
τ_0	relaxation time
ϕ	conductive absolute temperature
φ	$\phi - \varphi_0$; $\left\|\frac{\varphi}{\varphi_0}\right\| << 1$

1. Introduction

Chen and Gurtin [1] and Chen et al. [2,3] have formulated a theory of heat conduction in deformable bodies, which depends upon two distinct temperatures, the conductive temperature φ and thermodynamic temperature T. For infinitesimal theory, it is assumed that the conductive temperature deviations φ from reference temperature φ_o and the first two gradients of φ are small. Furthermore, the displacement $\boldsymbol{u}$, its first three gradients, and the velocity gradient are small. The linearized constitutive relations are *mechanically simple* in the sense that there is no dependence on $\nabla\mathbf{E}$ and $\nabla\nabla\mathbf{E}$, where $\mathbf{E}$ is the strain tensor [3].

In fact, if $a = 0$ or $\Delta\varphi = 0$ so that the two temperatures coincide, then the constitutive equations of the two-temperature thermoelasticity are just the classical equations of the linearized thermoelasticity. The two-temperature parameter $a > 0$ represents a *singular perturbation* on the field equations [4]. Iesan [5] established the uniqueness, reciprocity theorems, and variational principle for homogeneous isotropic solid in the frame of coupled thermoelasticity theory involving two temperatures. Youssef [6] investigated the two-temperature theory in the context of the generalized theory of thermoelasticity.

The propagation of harmonic plane waves in media described by the two-temperature theory of thermoelasticity has been investigated by Puri and Jordan [7], where they extended the work of Chakrabarti [8], derived the dispersion relation, and examined several measures of the wave field. Ezzat and El-Bary [9] established the model of one-dimensional equations of the two-temperature generalized magneto-thermo-viscoelasticity and adopting the state space approach developed by Ezzat [10] gave the solution for any set of boundary conditions. Existence, structural stability, convergence, and spatial behavior in two-temperature thermoelasticity theory were given by Quintanilla [11].

The two-temperature thermoelasticity theory (2TT) and the classical theory of thermoelasticity (CTE) suffer from the so-called paradox of heat conduction, i.e., the heat equations for both theories of a mixed parabolic–hyperbolic type, predicting infinite speeds of propagation for heat waves contrary to physical observations. The generalized thermoelasticity theories in which the heat conduction equation is hyperbolic do not suffer from this paradox.

Several generalizations to the coupled theory are introduced. One can refer to Ignaczak [12] and to Chandrasekharaiah [13] for a review, presentation of generalized theories. Hetnarski and Ignaczak in their

survey article [14] examined five generalizations to the coupled theory and obtained a number of important analytical results. Hetnarski and Eslami [15] introduced a unified generalized thermoelasticity theory and present advanced theory and applications of classical thermoelasticity, generalized thermoelasticity, and mathematical and mechanical background of thermodynamics and theory of elasticity as well.

The mathematical aspects of Lord-Shulman [16] theory are explained and illustrated in detail in the work of Ignaczak and Ostoja-starzeweski [17]. Joseph and Preziosi [18,19] state that the Cattaneo heat conduction law [20] is the most obvious and simple generalization of the Fourier law that gives rise to a finite propagation speed.

Within the theoretical contributions to the subject are the proofs of uniqueness theorems under different conditions by Ignaczak [21], Chandrasekharaiah [22], Sherief [23], and Ezzat and El Karamany [24,25]. The fundamental solutions for generalized thermoelasticity problem were obtained by Sherief [26] and Ezzat [27]. The propagation of discontinuities of solutions in this theory was investigated in Ref. [28,29].

Differential equations of fractional order have been the focus of many studies due to their frequent appearance in various applications in fluid mechanics, viscoelasticity, biology, physics, and engineering. The most important advantage of using fractional differential equations in these and other applications is their non-local property. It is well known that the integer order differential operator is a local operator but the fractional order differential operator is non-local. This means that the next state of a system depends not only upon its current state but also upon all of its historical states. This is more realistic, and it is one reason why fractional calculus has become more and more popular [30–32].

Although the tools of fractional calculus were available and applicable to various fields of study, the investigation into the theory of fractional differential equations started quite recently [30]. The differential equations involving Riemann–Liouville differential operators of fractional order $0 < \alpha < 1$ appear to be important in modeling several physical phenomena [33] and therefore seem to deserve an independent study of their theory parallel to the well-known theory of ordinary differential equations.

Fractional calculus has been used successfully to modify many existing models of physical processes. One can state that the whole theory of fractional derivatives and integrals was established in the second half of the nineteenth century. The first application of fractional derivatives was given by Abel who applied fractional calculus in the solution of an integral equation that arises in the formulation of the Tautochrone problem. The generalization of the concept of derivative and integral to a non-integer order has been subjected to several approaches, and some various alternative definitions of fractional derivatives appeared in Refs. [34–38]. In the last few years, fractional calculus was applied successfully in various areas to modify many existing models of physical processes, e.g., chemistry, biology, modeling and identification, electronics, wave propagation, and viscoelasticity [39–43]. One can refer to Padlubny [32] for a survey of applications of fractional calculus.

Recently, a considerable research effort is expended to study anomalous diffusion, which is characterized by the time-fractional diffusion-wave equation by Kimmich [44]

$$\rho C = \lambda I^{\upsilon} \nabla^2 C, \quad 0 < \upsilon \leq 2, \tag{1}$$

where the notion I^{υ} is the Riemann–Liouville fractional integral, which was introduced as a natural generalization of the well-known n-fold repeated integral $I^n f(t)$ written in a convolution-type form [45]

$$\left.\begin{array}{l} I^{\upsilon} f(t) = \frac{1}{\Gamma(\upsilon)} \int\limits_0^t (t-\xi)^{\upsilon-1} f(\xi) d\xi \\ I^0 f(t) = f(t) \end{array}\right\} \qquad 0 < \upsilon \leq 2. \tag{2}$$

According to Kimmich [44], Eq. (1) describes different cases of diffusion where $0 < \upsilon < 1$ corresponds to weak diffusion (subdiffusion), $\upsilon = 1$ corresponds to normal diffusion, $0 < \upsilon < 2$ corresponds to strong diffusion (superdiffusion), and $\upsilon = 2$ corresponds to ballistic diffusion. It should be noted that the term diffusion is often used in a more generalized sense including various transport phenomena. Equation (1)

is a mathematical model of a wide range of important physical phenomena; for example, the subdiffusive transport occurs in widely different systems ranging from dielectrics and semiconductors through polymers to fractals, glasses, porous, and random media. Superdiffusion is comparatively rare and has been observed in porous glasses, polymer chain, biological systems, transport of organic molecules and atomic clusters on surface. One might expect the anomalous heat conduction in media where the anomalous diffusion is observed.

Fujita [46,47] considered the constitutive equation for the heat flux in the form

$$\boldsymbol{q} = -\kappa I^{\upsilon-1}\nabla T, \quad 1 < \upsilon \leq 2. \tag{3}$$

Povstenko [48] used the Caputo heat wave equation to define the fractional heat conduction equation in the form

$$\boldsymbol{q} = -\kappa I^{\upsilon-1}\nabla T, \quad 0 < \upsilon \leq 2, \tag{4}$$

Cattaneo [20] introduced a law of heat conduction to replace the classical Fourier law in the form

$$\boldsymbol{q} + \tau_o \frac{\partial \boldsymbol{q}}{\partial t} = -\kappa \nabla T. \tag{5}$$

Sherief et al. [49] introduced a formula of heat conduction and taking into account consideration (7)

$$\boldsymbol{q} + \tau_o \frac{\partial^{\alpha} \boldsymbol{q}}{\partial t^{\alpha}} = -\kappa \nabla T, \qquad 0 < \alpha \leq 1, \tag{6}$$

where

$$\frac{\partial^{\alpha}}{\partial t^{\alpha}} f(y,t) = \begin{cases} f(y,t) - f(y,0) & \alpha \to 0 \\ I^{\alpha-1} \frac{\partial f(y,t)}{\partial t} & 0 < \alpha < 1, \\ \frac{\partial f(y,t)}{\partial t} & \alpha = 1 \end{cases} \tag{7}$$

and proved a uniqueness theorem and derived a reciprocity relation and a variational principle.

In the limit, as α tends to one, Eq. (6) reduces to the well-known Cattaneo law used by Lord and Shulman [16] to derive the equation of the generalized theory of thermoelasticity with one relaxation time. It is known that Lebon et al. [50] and Jou et al. [51] show that the classical entropy derived using this law instead of being monotonically increasing behaves in an oscillatory way. Strictly speaking, this result is not incompatible with the Clausius' formulation of the second law, which states that the entropy of the final equilibrium state must be higher than the entropy of the initial equilibrium state. However, the non-monotonic behavior of the entropy is in contradiction with the local equilibrium formulation of the second law, which requires that the entropy production must be positive everywhere at any time [50]. During the last two decades, this became the subject of many research papers and resulted in the introduction of what is known now as extended irreversible thermodynamics. A review can be found in Jou et al. [51].

Youssef [52] introduced another formula of heat conduction and taking into consideration (3)–(5)

$$\boldsymbol{q} + \tau_o \frac{\partial \boldsymbol{q}}{\partial t} = -kI^{\upsilon-1}\nabla T, \quad 0 < \upsilon \leq 2, \tag{8}$$

and a uniqueness theorem has been proved.

Ezzat [53,54] established a new model of fractional heat conduction equation by using the new Taylor series expansion of time-fractional order, developed by Jumarie [55] as

$$\boldsymbol{q} + \frac{\tau_o^{\alpha}}{\alpha!} \frac{\partial^{\alpha} \boldsymbol{q}}{\partial t^{\alpha}} = -k\nabla T, \quad 0 < \alpha \leq 1. \tag{9}$$

El-Karamany and Ezzat [56] introduced two general models of fractional heat conduction law for a non-homogeneous anisotropic elastic solid. Uniqueness and reciprocal theorems are proved, and the convolutional variational principle is established and used to prove a uniqueness theorem with no restriction on the elasticity or thermal conductivity tensors except symmetry conditions. The two-temperature

dynamic coupled, Lord-Shulman and fractional coupled thermoelasticity theories result as limit cases. For fractional thermoelasticity not involving two temperatures, El-Karamany and Ezzat [57] established the uniqueness, reciprocal theorems and convolution variational principle. The dynamic coupled and Green–Naghdi thermoelasticity theories result as limit cases. The reciprocity relation in case of quiescent initial state is found to be independent of the order of differintegration [56] and [57]. Fractional order theory of a perfect conducting thermoelastic medium not involving two temperatures was investigated by Ezzat and El-Karamany [58]

In the current work, a new model of time-fractional derivative of order α in heat conduction equation has been derived in the context of generalized thermoelectric elasticity theory. The governing coupled equations are applied to a problem of a perfect conducting half-space with heat source distribution in the presence of a transverse magnetic field. Laplace transforms and state space approach techniques are used to get the solution. Laplace transforms are obtained using the complex inversion formula of the transform together with Fourier expansion techniques proposed by Honig and Hirdes [59]. The effects of various physical parameters on various heat transfer, stress, displacement and strain characteristics are discussed in detail and represented graphically.

2. Fractional heat conduction equation of two temperatures thermoelasticity

The conventional electro-thermoelasticity is based on the principles of the classical theory of heat conductivity, specifically on the classical Fourier's law, in which relates the heat flux vector $\boldsymbol{q}$ to the temperature gradient

$$\boldsymbol{q} = -k\nabla\theta. \tag{10}$$

The energy equation in terms of the heat flux vector $\boldsymbol{q}$, Biot [60]

$$\frac{\partial}{\partial t}\left(\rho C_E\theta + \gamma T_0 e\right) = -\nabla.\boldsymbol{q} + Q. \tag{11}$$

During the past three decades, non-classical electro-thermoelasticity theories, in which Fourier law (10) and heat equation (11) are replaced by more general equations, have been formulated by taking Taylor's series to expand $\boldsymbol{q}(\boldsymbol{x},\ t+\tau_o)$ and retaining terms up to the first order in τ_o. The first well-known generalization of such a type

$$\boldsymbol{q} + \tau_o\frac{\partial \boldsymbol{q}}{\partial t} = -k\nabla\theta, \tag{12}$$

leads to the hyperbolic-type heat transport equation in the theory of generalized thermoelasticity [16]

$$\frac{\partial}{\partial t}\left(1+\tau_o\frac{\partial}{\partial t}\right)\left(\rho C_E\theta + \gamma T_0 e\right) = k\nabla^2\theta + Q + \tau_o\frac{\partial Q}{\partial t}. \tag{13}$$

Recently, Youssef [6] investigated two-temperature generalized thermoelasticity theory in which Fourier law (10) is replaced by

$$\boldsymbol{q} + \tau_o\frac{\partial \boldsymbol{q}}{\partial t} = -k\nabla\varphi, \tag{14}$$

$$\phi - T = \varphi - \theta = a\nabla^2\varphi, \tag{15}$$

leads to the heat equation in the form

$$\frac{\partial}{\partial t}\left(1+\tau_o\frac{\partial}{\partial t}\right)\left(\rho C_E\theta + \gamma\varphi_0 e\right) = k\nabla^2\varphi + Q + \tau_o\frac{\partial Q}{\partial t}. \tag{16}$$

In this paper, the new fractional Taylor's series of time-fractional order α developed in Jumarie [56] is adopted to expand $\boldsymbol{q}(\boldsymbol{x},\ t+\tau_o)$ and retaining terms up to order α in the thermal relaxation time τ_o, we get

$$\boldsymbol{q}(\boldsymbol{x},\ t+\tau_o)=\boldsymbol{q}(\boldsymbol{x},\ t)+\frac{\tau_o^{\alpha}}{\alpha!}\frac{\partial^{\alpha}\boldsymbol{q}}{\partial t^{\alpha}},\quad 0<\alpha\leq 1. \tag{17}$$

From a mathematical viewpoint, Fourier law (14) in the theory of generalized fractional heat conduction involving two temperatures is given by

$$\boldsymbol{q}+\frac{\tau_o^{\alpha}}{\alpha!}\frac{\partial^{\alpha}\boldsymbol{q}}{\partial t^{\alpha}}=-k\nabla\varphi,\quad 0<\alpha\leq 1. \tag{18}$$

Taking the partial time-derivative of fraction order α of Eq. (11), we get

$$\frac{\partial^{\alpha+1}}{\partial t^{\alpha+1}}\left(\rho C_E\theta+\gamma\varphi_0 e\right)=-\nabla.\left(\frac{\partial^{\alpha}\boldsymbol{q}}{\partial t^{\alpha}}\right)+\frac{\partial^{\alpha}\boldsymbol{Q}}{\partial t^{\alpha}},\qquad 0<\alpha\leq 1. \tag{18}$$

Multiplying Eq. (18) by $\frac{\tau_o^{\alpha}}{\alpha!}$ and adding to Eq. (11), we have

$$\frac{\partial}{\partial t}\left(1+\frac{\tau_o^{\alpha}}{\alpha!}\frac{\partial^{\alpha}}{\partial t^{\alpha}}\right)\left(\rho C_E\theta+\gamma\varphi_o e\right)=-\nabla.\left(\boldsymbol{q}+\frac{\tau_o^{\alpha}}{\alpha!}\frac{\partial^{\alpha}\boldsymbol{q}}{\partial t^{\alpha}}\right)+Q+\frac{\tau_o^{\alpha}}{\alpha!}\frac{\partial^{\alpha}Q}{\partial t^{\alpha}},\quad 0<\alpha\leq 1, \tag{19}$$

Substituting from Eq. (18), we get

$$\frac{\partial}{\partial t}\left(1+\frac{\tau_o^{\alpha}}{\alpha!}\frac{\partial^{\alpha}}{\partial t^{\alpha}}\right)\left(\rho C_E\theta+\gamma\varphi_o e\right)=k\nabla^2\varphi+\left(1+\frac{\tau_o^{\alpha}}{\alpha!}\frac{\partial^{\alpha}}{\partial t^{\alpha}}\right)Q,\quad 0<\alpha\leq 1. \tag{20}$$

Equation (20) is the generalized energy equation with fractional time derivatives in which the relaxation time τ_o is considered. Some theories of heat conduction law follow as limit cases for different values of the parameters α and τ_o.

Limiting cases

(i) In the theory of thermoelasticity
1. The heat Eq. (20) in the limiting case $\theta=\varphi$ and $\tau_o=0$ transforms to the work of Biot [60].
2. The heat Eq. (20) in the limiting case $\theta=\varphi,\ \tau_o=0$ and $\upsilon=1$ transforms to the work of Povstenko [48]

(ii) In the theory of two-temperature thermoelasticity
3. The heat Eq. (20) in the limiting case $\theta\neq\varphi$ and $\tau_o=0$ transforms to the work of Chen and Gurtin [1] and Iesan [5].

(iii) In the theory of generalized thermoelasticity
4. The heat Eq. (20) in the limiting case $\theta=\varphi,\ \tau_o>0$ and $\alpha=1$ transforms to the work of Lord and Shulman [16].
5. The heat Eq. (20) in the limiting case $\theta=\varphi$ and $0<\alpha\leq 1$ transforms to the work of Sherief et al. [49] and Ezzat [54].
6. The heat Eq. (20) in the limiting case $\theta=\varphi,\alpha=1$ and $\upsilon=1$ transforms to the work of Youssef [52].

(iv) In the theory of two-temperature generalized thermoelasticity
7. The heat Eq. (20) in the limiting case $\theta\neq\varphi,\ \alpha=1$ and $\tau_o>0$ transforms to the work of Youssef [6], Ezzat and El-Bary [9], Ezzat and El-Karamany [61] and Ezzat et al. [62].

3. The physical problem and state space approach

We shall consider a thermoelastic medium of prefect conductivity permeated by an initial magnetic field $\boldsymbol{H}$. This produces an induced magnetic field $\boldsymbol{h}$ and induced electric field $\boldsymbol{E}$, which satisfy the linearized equations of electromagnetism and are valid for slowly moving media. The governing equations for generalized two-temperature magneto-thermoelasticity consist of [58]

$$\operatorname{curl} \boldsymbol{h} = \boldsymbol{J} + \varepsilon_o \frac{\partial \boldsymbol{E}}{\partial t}, \tag{21}$$

$$\operatorname{curl} \boldsymbol{E} = -\mu_o \frac{\partial \boldsymbol{h}}{\partial t}, \tag{22}$$

$$\boldsymbol{E} = -\mu_o \left(\frac{\partial \boldsymbol{u}}{\partial t} \wedge \boldsymbol{H} \right), \tag{23}$$

$$\operatorname{div} \boldsymbol{h} = 0. \tag{24}$$

The equation of motion in the absence of body forces

$$\sigma_{ji,j} + \mu_o (J \wedge H)_i = \rho \frac{\partial^2 u_i}{\partial t^2}, \tag{25}$$

The heat equation with fractional time derivatives

$$\left(1 + \frac{\tau_o^\alpha}{\alpha!} \frac{\partial^\alpha}{\partial t^\alpha}\right) \left(\rho C_E \frac{\partial \theta}{\partial t} + \varphi_o \gamma \frac{\partial e}{\partial t} - Q\right) = k\varphi_{,ii}, \quad 0 < \alpha \leq 1. \tag{26}$$

The relation between the heat conduction and dynamical heat

$$\varphi - \theta = a\varphi_{,ii} \tag{27}$$

where a > 0, is the temperature discrepancy.

The constitutive equation

$$\sigma_{ij} = 2\mu e_{ij} \ + \ \lambda e \delta_{ij} \ - \ \gamma \theta \delta_{ij}. \tag{28}$$

The strain–displacement relations

$$e_{ij} = \frac{1}{2} \left(u_{i.j} + u_{j,i}\right). \tag{29}$$

In the above equations, a comma denotes material derivatives and the summation convention are used.

Now, we shall consider a homogeneous isotropic thermoelastic perfect conducting solid occupying half-space $x \geq 0$, which is initially quiescent and where all the state functions depend only on the state functions are depend only on the dimension x and the time t and the displacement vector has components $(u(x,t), 0, 0)$. A constant magnetic field with components $(0, H_o, 0)$ is permeating the medium. The induced magnetic field $\boldsymbol{h}$ will have one component in the y-direction, while the induced electric field $\boldsymbol{E}$ will have one component in z-direction.

Let us introduce the following non-dimensional variables:

$$x' = c_o \eta_o x, \quad u' = c_o \eta_o u, \quad t' = c_o^2 \eta_o t, \quad \tau_0' = c_o^2 \eta_o \tau_o, \quad \theta' = \frac{\gamma \theta}{\rho c_o^2}, \quad \sigma_{ij}' = \frac{\sigma_{ij}}{\rho c_o^2},$$

$$q_i' = \frac{q_i \gamma}{k \rho c_o^3 \eta_o}, \quad h' = \frac{h}{H_o}, \quad E' = \frac{E}{\mu_o H_o c_o}, \quad \varphi' = \frac{\gamma \varphi}{\rho c_o^2}, \quad \eta_o = \frac{\rho C_E}{\kappa}, \quad \varepsilon = \frac{\varphi_0 \gamma^2}{k \rho c_o^2 \eta_o},$$

$$Q' = \frac{Q\gamma}{k \rho c_0^4 \eta_0^2}$$

The dimensionless temperature discrepancy is $\beta_o = a c_o^2 \eta_o^2 = a(c_o/k)^2$ [4].

Using homogeneity and scale change of fractional derivatives [34], the following system of equations in terms of the preceding non-dimensional variables result (suppressing the primes for convenience)

$$h = -e, \tag{30}$$

$$E = -\frac{\partial u}{\partial t}, \tag{31}$$

$$\frac{\partial^2 \varphi}{\partial x^2} = \left(\frac{\partial}{\partial t} + \frac{\tau_o^\alpha}{\alpha!}\frac{\partial^{\alpha+1}}{\partial t^{\alpha+1}}\right)(\theta + \varepsilon e) - \left(1 + \frac{\tau_o^\alpha}{\alpha!}\frac{\partial^\alpha}{\partial t^\alpha}\right) Q, \tag{32}$$

$$\frac{\partial^2 \sigma}{\partial x^2} + \beta \frac{\partial^2 e}{\partial x^2} = \alpha^* \ddot{e}, \tag{33}$$

$$\sigma = e - \theta, \tag{34}$$

$$\varphi - \theta = \beta_o \frac{\partial^2 \varphi}{\partial x^2}, \tag{35}$$

where $\alpha^* = 1 + \frac{\alpha_o^2}{c^2}$ and $\beta = (\alpha_o/c_o)^2$.

From now on, we shall consider a heat source of the form $Q = Q_o \delta(x) H(t)$, where Q_0 is a positive constant.

To simplify the algebra, only problems with zero initial conditions are considered. Applying the Laplace transform defined by the formulas [48]

$$\left.\begin{array}{ll} L\{g(t)\} = \bar{g}(s) = \int_0^\infty \mathrm{e}^{-st} g(t)\, \mathrm{d}t & \\ L\{D^{\mathrm{n}} g(t)\} = s^{\mathrm{n}} L\{g(t)\} & n > 0 \end{array}\right\}, \tag{36}$$

on both sides of Eqs. (30)–(35) and writing the resulting equations in matrix form results in

$$\frac{\mathrm{d}^2}{\mathrm{d}x^2}\left\{\begin{array}{l} \bar{\varphi} \\ \bar{\sigma} \end{array}\right\} = \left\{\begin{array}{ll} L_1 & L_2 \\ M_1 & M_2 \end{array}\right\}\left\{\begin{array}{l} \bar{\varphi} \\ \bar{\sigma} \end{array}\right\} - Q_o \beta^* \delta(x) \left\{\begin{array}{l} 1 \\ 0 \end{array}\right\}, \tag{37}$$

where

$$\varpi = s\left(1 + \frac{\tau_o^\alpha}{\alpha!} s^\alpha\right), \quad \beta^* = \frac{\varpi}{s\left[1 + \beta_o \varpi (1+\varepsilon)\right]}, \quad L_1 = \frac{\varpi(1+\varepsilon)}{1 + \beta_o \varpi (1+\varepsilon)}, \quad L_2 = \frac{\varpi \varepsilon}{1 + \beta_o \varpi (1+\varepsilon)}$$

$$M_1 = \frac{m\left(\alpha s^2 - \beta L_1\right)}{1 + \beta n}, \quad M_2 = \frac{n\left(\alpha s^2 - \beta L_2\right)}{1 + \beta n}, \quad m = 1 - \beta_o L_1 \text{ and } n = 1 - \beta_o L_2.$$

Choosing as state variables the temperature of heat conduction $\bar{\varphi}$ and the stress component $\bar{\sigma}$ in the x-direction, Eq. (37) can be written in the absence of heat source as

$$\frac{\mathrm{d}^2 \bar{G}(x,s)}{\mathrm{d}x^2} = A(s)\bar{G}(x,s), \tag{38}$$

where

$$\bar{G}(x,s) = \begin{bmatrix} \bar{\varphi}(x,s) \\ \bar{\sigma}(x,s) \end{bmatrix} \text{ and } A(s) = \begin{bmatrix} L_1 & L_2 \\ M_1 & M_2 \end{bmatrix}.$$

The formal solution of system (37) can be written in the form

$$\bar{G}(x,s) = \exp\left[-\sqrt{A(s)}x\right]\bar{G}(0,s), \tag{39}$$

where

$$\bar{G}(0,s) = \begin{bmatrix} \bar{\varphi}(0,s) \\ \bar{\sigma}(0,s) \end{bmatrix} = \begin{bmatrix} \bar{\varphi}_o \\ \bar{\sigma}_o \end{bmatrix}$$

where for bounded solution with large x, we have canceled the part of exponential that has a positive power.

We shall use the well-known Cayley–Hamilton theorem to find the form of the matrix $\exp[-\sqrt{A(s)}x]$. The characteristic equation of the matrix $A(s)$ can be written as follows:

$$k^2 - k(L_1 + M_2) + (L_1 M_2 - L_2 M_1) = 0. \tag{40}$$

The roots of this equation, namely k_1 and k_2, satisfy the following relations:

$$k_1 + k_2 = L_1 + M_2 \tag{41a}$$

$$k_1 k_2 = L_1 M_2 - L_2 M_1. \tag{41b}$$

The Taylor series expansion of the matrix exponential in Eq. (39) has the form

$$\exp\left[-\sqrt{A(s)}x\right] = \sum_{n=0}^{\infty} \frac{[-\sqrt{A(s)}x]^n}{n!}. \tag{42}$$

Using the Cayley–Hamilton theorem, we can express A^2 and higher powers of the matrix A in terms of I and A, where I is the unit matrix of second order.

Thus, the infinite series in equation (42) can be reduced to

$$\exp\left[-\sqrt{A(s)}x\right] = a_o(x,s)I + a_1(x,s)A(s). \tag{43}$$

where a_0 and a_1 are coefficients depending on x and s.

By the Cayley–Hamilton theorem, the characteristic roots k_1 and k_2 of the matrix A must satisfy equation (42), thus

$$\exp\left[-\sqrt{k_1}x\right] = a_o + a_1 k_1, \tag{44a}$$

$$\exp\left[-\sqrt{k_2}x\right] = a_o + a_1 k_2. \tag{44b}$$

The solution of the above system is given by

$$a_o = \frac{k_1 \mathrm{e}^{-\sqrt{k_2}x} - k_2 \mathrm{e}^{-\sqrt{k_1}x}}{k_1 - k_2}, \quad \text{and} \quad a_1 = \frac{\mathrm{e}^{-\sqrt{k_1}x} - \mathrm{e}^{-\sqrt{k_2}x}}{k_1 - k_2}.$$

Hence, we have

$$\exp\left[-\sqrt{k_1}x\right] = L_{ij}(x,s), \quad i,j = 1,2,$$

where

$$\begin{aligned} L_{11} &= \frac{\mathrm{e}^{-\sqrt{k_2}x}(k_1 - L_1) - \mathrm{e}^{-\sqrt{k_1}x}(k_2 - L_1)}{k_1 - k_2}, \quad L_{12} = \frac{L_2\ \left(\mathrm{e}^{-\sqrt{k_1}x} - \mathrm{e}^{-\sqrt{k_2}x}\right)}{k_1 - k_2} \\ L_{22} &= \frac{\mathrm{e}^{-\sqrt{k_1}x}(k_2 - M_2) - \mathrm{e}^{-\sqrt{k_2}x}(k_1 - M_2)}{k_1 - k_2}, \quad L_{21} = \frac{M_1\ \left(\mathrm{e}^{-\sqrt{k_1}x} - \mathrm{e}^{-\sqrt{k_2}x}\right)}{k_1 - k_2}. \end{aligned} \tag{45}$$

The solution in Eq. (39) can be written in the form

$$\bar{G}(x,s) = L_{ij}\ \bar{G}(0,s). \tag{46}$$

Hence, we obtain

$$\bar{\varphi}(x,s) = \frac{(k_1\bar{\varphi}_o - L_1\bar{\varphi}_o - L_2\bar{\sigma}_o)\ \mathrm{e}^{-\sqrt{k_2}x} - \ (k_2\bar{\varphi}_o - L_1\bar{\varphi}_o - L_2\bar{\sigma}_o)\ \mathrm{e}^{-\sqrt{k_1}x}}{k_1\ -\ k_2}, \tag{47}$$

$$\bar{\sigma}(x,s) = \frac{(k_1\bar{\sigma}_o - M_1\bar{\varphi}_o - M_2\bar{\sigma}_o)\mathrm{e}^{-\sqrt{k_2}x} - \ (k_2\bar{\sigma}_o - M_1\bar{\varphi}_o - M_2\bar{\sigma}_o)\mathrm{e}^{-\sqrt{k_1}x}}{k_1\ -\ k_2}. \tag{48}$$

By using Eqs. (47) and (48) with Eq. (35) we get

$$\bar{\Theta}(x,s)=\frac{(k_1\bar{\varphi}_o-L_1\bar{\varphi}_o-L_2\bar{\sigma}_o)(1-\beta_o k_2)\mathrm{e}^{-\sqrt{k_2}x}-(k_2\bar{\varphi}_o-L_1\bar{\varphi}_o-L_2\bar{\sigma}_o)(1-\beta_o k_1)\mathrm{e}^{-\sqrt{k_1}x}}{k_1-k_2} \tag{49}$$

It should be noted that the corresponding expressions for two-temperature generalized thermoelasticity with relaxation time in the absence of magnetic field can be deduced by setting $\alpha=1$ and $\beta=0$ in Eq. (45).

We consider a semi-space homogeneous thermoelastic medium of perfect conductivity occupying the region $x\geq 0$ with quiescent initial state and boundary conditions in the following form:

(i) Thermal boundary condition:

A thermal shock is applied to the boundary plane $x=0$ in the form

$$\varphi(0,t)=\varphi_o H(t),\quad \text{or}\quad \bar{\varphi}(0,s)=\bar{\varphi}_o=\frac{\varphi_o}{s} \tag{50}$$

where $\bar{\varphi}_o$ is a constant and $H(t)$ is the Heaviside unit step function.

(ii) Mechanical boundary condition:

The bounding plane $x=0$ is taken to be traction-free, i.e.,

$$\sigma(0,t)+T_{11}(0,t)-T_{11}^{\mathrm{o}}(0,t)=0, \tag{51}$$

where T_{11}^{o} is the Maxwell stress tensor in a vacuum.

Since the transverse components of the vectors E and h are continuous across the bounding plane, i.e., $E(0,t)=E^0(0,t)$ and $h(0,t)=h^0(0,t)$, $t>0$, where E^0 and h^0 are the components of the induced electric and magnetic field in free space and the relative permeability is very nearly unity, it follows that $T_{11}(0,t)=T_{11}^{\mathrm{o}}(0,t)$ and equation (51) reduces to [19]:

$$\sigma(0,t)=0,\quad \text{or}\quad \bar{\sigma}(0,s)=\bar{\sigma}_0=0. \tag{52}$$

Hence, we can use the conditions on (50–51) into Eqs. (47–48) to get the exact solution in the Laplace transform domain in the following forms:

$$\bar{\varphi}(x,s)=\frac{\varphi_o\left[(k_1-L_1)\,\mathrm{e}^{-\sqrt{k_2}x}-(k_2-L_1)\,\mathrm{e}^{-\sqrt{k_1}x}\right]}{s(k_1-k_2)}, \tag{53}$$

$$\bar{\sigma}(x,s)=\frac{\varphi_o M_1\left(\mathrm{e}^{-\sqrt{k_1}x}-\mathrm{e}^{-\sqrt{k_2}x}\right)}{s(k_1-k_2)}, \tag{54}$$

$$\bar{\Theta}(x,s)=\frac{\varphi_o\left[B\,\mathrm{e}^{-\sqrt{k_2}x}-A\,\mathrm{e}^{-\sqrt{k_1}x}\right]}{s(k_1-k_2)}, \tag{55}$$

$$\bar{e}(x,s)=\frac{\varphi_o\left[[m(k_1-L_1)-nM_1]\,\mathrm{e}^{-\sqrt{k_2}x}-[m(k_2-L_1)-nM_1]\,\mathrm{e}^{-\sqrt{k_1}x}\right]}{s(k_1-k_2)}, \tag{56}$$

where $A=(k_2-L_1)(1-\beta_o k_1), B=(k_1-L_1)(1-\beta_o k_2)$.

From Eq. (29), the displacement takes the form:

$$\bar{u}(x,s)=\frac{\varphi_o\left(C\mathrm{e}^{-\sqrt{k_1}x}-D\mathrm{e}^{-\sqrt{k_2}x}\right)}{s(k_1-k_2)}, \tag{57}$$

where $C=\frac{m(k_2-L_1)-nM_1}{\sqrt{k_1}}$, $D=\frac{m(k_1-L_1)-nM_1}{\sqrt{k_2}}$.

The induced magnetic and electric field takes the following forms

$$\bar{h}(x,s) = -\frac{\varphi_o\left[[m(k_1-L_1)-nM_1]\,\mathrm{e}^{-\sqrt{k_2}x} - [m(k_2-L_1)- nM_1]\,\mathrm{e}^{-\sqrt{k_1}x}\right]}{s(k_1 - k_2)}, \tag{58}$$

$$\bar{E}(x,s) = -\frac{\varphi_o\left(C\mathrm{e}^{-\sqrt{k_1}x} - D\mathrm{e}^{-\sqrt{k_2}x}\right)}{s^2(k_1 - k_2)}. \tag{59}$$

Those complete the solution in the Laplace transform domain.

4. Numerical inversion of the laplace transforms

In order to invert the Laplace transform in the above equations, we adopt a numerical inversion method based on a Fourier series expansion proposed [56]. In this method, the inverse $g(t)$ of the Laplace transform $\bar{g}(s)$ is approximated by the relation

$$g(t) = \frac{\mathrm{e}^{c^*t}}{t_1}\left[\frac{1}{2}\bar{g}(c^*) + Re\left(\sum_{k=1}^{N}\exp\left(\frac{ik\pi t}{t_1}\right)\bar{g}\left(c^* + \frac{ik\pi}{t_1}\right)\right)\right], \quad 0 \le t \le 2t_1, \tag{60}$$

where c^* is an arbitrary constant greater than all the real parts of the singularities of $g(t)$ and N is sufficiently large integer chosen such that,

$$\mathrm{e}^{c^*t}Re\left[\exp\left(\frac{iN\pi t}{t_1}\right)\bar{g}\left(c^* + \frac{iN\pi}{t_1}\right)\right] \le \varepsilon, \tag{61}$$

where ε is a prescribed small positive number that corresponds to the degree of accuracy required.

Using the numerical procedure cited, to invert the expressions of temperature, stresses and displacement fields in Laplace transform domain.

5. Numerical results and discussion

The copper material was chosen for purposes of numerical evaluations. The constants of the problem were taken as in Table 1 [61].

The investigation into the effect of the fractional order α on perfect conducting thermoelastic material in the presence of a magnetic field has been carried out in the preceding sections. The computations were performed for a value of time, namely $t = 0.1$. The numerical technique outlined earlier was used to obtain the conductive temperature, the thermodynamic temperature, the stress, the displacement and the strain as well as the induced magnetic field and the induced electric field distributions. The results are represented graphically at different positions of x as shown in Figs. 1, 2, 3, 4, 5. In these figures, we noticed the difference in all functions for the value of $\alpha(0 < \alpha \le 1)$ where the case of $\alpha = 1$ (normal conductivity) indicates the old situation and the case $0 < \alpha < 1$ (weak conductivity) indicates the new theory. For a normal conductivity $\alpha = 1$, the results coincide with all the previous results of applications that are taken in the context of the generalized thermoelasticity with one relaxation time in on the various fields [9,61].

Table 1. *Values of the constants*

$k = 386$ N/Ks, $\alpha_T = 1.78 \times 10^{-5}$ K^{-1}, $C_E = 383.1$ m^2/K, $\eta_o = 8886.73$ s/m^2,
$\mu = 3.86 \times 10^{10}$ N/m^2, $\lambda = 7.76 \times 10^{10}$ N/m^2, $\rho = 8954$ kg/m^3,
$c_o = 4{,}158$ m/s, $T_o = 293$ K, $\varepsilon = 0.0168$, $\tau_o = 0.02$ s.

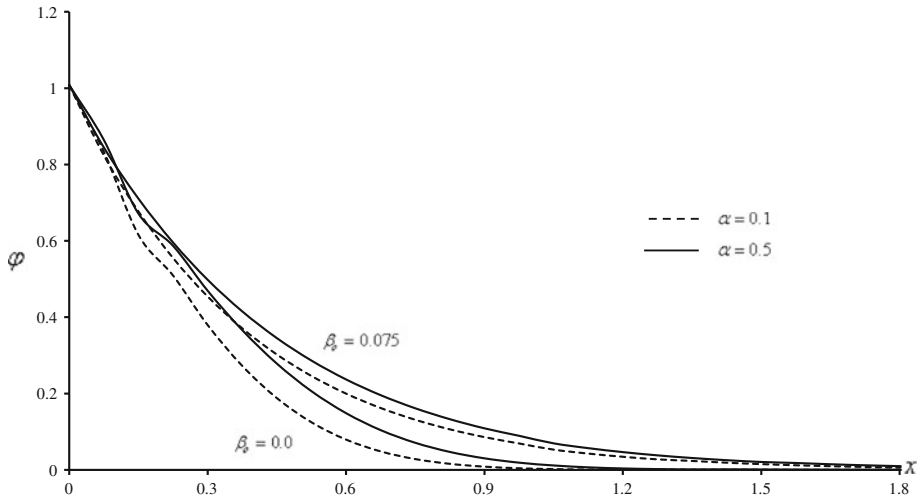


FIG. 1. The conductive temperature distribution in two theories for different values of fractional parameter α

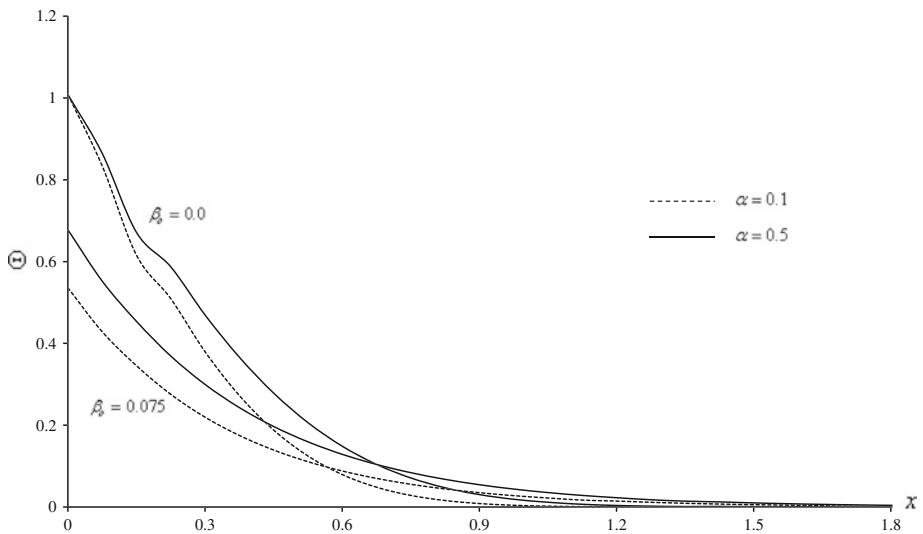


FIG. 2. The thermodynamic temperature distribution in two theories for different values of fractional parameter α

In Figs. 1, 2, 3, 4, 5 exhibiting the space variation of conductive temperature, thermodynamic temperature, stress, displacement and strain fields at different values of α for two cases of the non-dimensional temperature discrepancy β_o where the case of $\beta_o = 0.0$ indicates the old situation (one temperature) and the case $\beta_o = 0.075$ [7] indicates the two-temperature theory, we observe the following:

- The fields are continuous functions for different values of $\alpha(0 < \alpha < 1)$ and are smoother in the case $\beta_o = 0.075$.
- The fractional order α has a significant effect on all fields.
- The waves reach the steady state depending on the value of the fractional orders α and temperature discrepancy β_o.

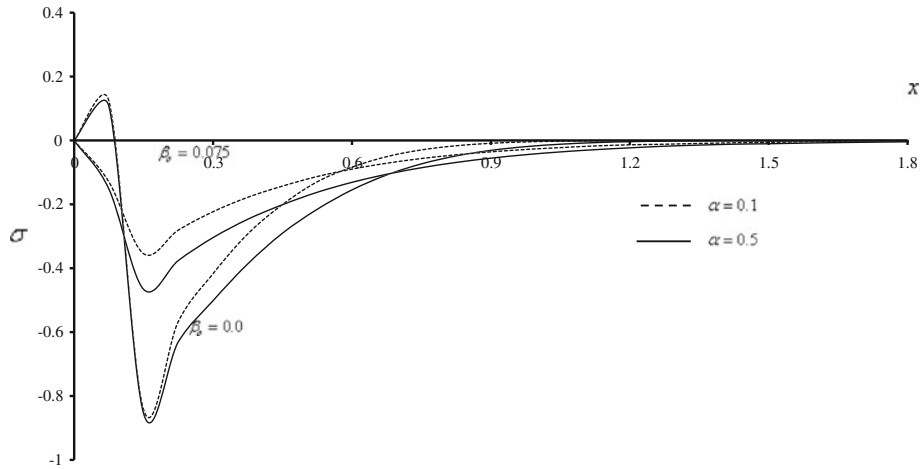


FIG. 3. The stress component distribution in two theories for different values of fractional parameter α

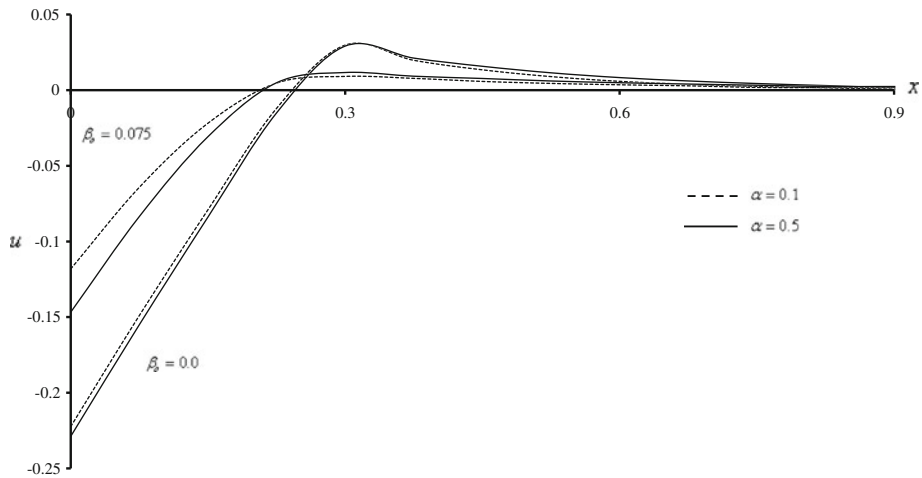


FIG. 4. The displacement component distribution in two theories for different values of fractional parameter α

- The waves cut the x-axis rapidly when $\alpha = 1, \beta_o = 0.0$ than when $0 < \alpha < 1$, and $\beta_o = 0.075$.
- Figures 1 and 2 display the conductive temperature and thermodynamic temperature distributions for wide range of $0 \leq x \leq 1.8$ at value of time $t = 0.1$ and for different values of the differential fractional order $\alpha(0 < \alpha < 1)$ and temperature discrepancy β_o. We have noticed that increasing the value of the parameter α decreases the conductive temperature and thermodynamic temperature, while the decreasing the value of the parameter β_o decreases the conductive temperature and increase in thermodynamic temperature.
- Figures 3, 4 and 5 present the stress, displacement, and strain fields which have the same behavior as the thermodynamic temperature distributions except the wide range of x.

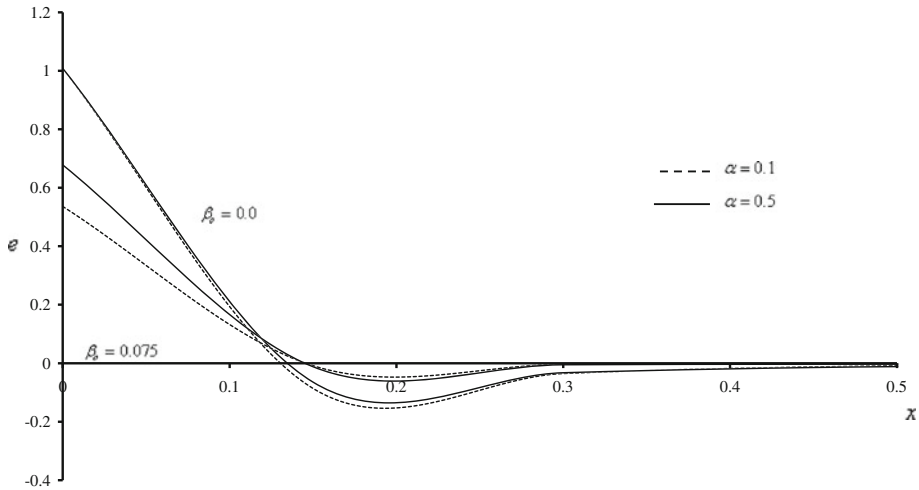


FIG. 5. The strain distribution in two theories for different values of fractional parameter α

6. Concluding remarks

- The main goal of this work is to introduce a new mathematical model for Fourier law of heat conduction with time-fractional order derivative. This model enables us to improve the efficiency of a thermoelectric material figure-of-merit. For a material to be good thermoelectric cooler, it must have a high thermoelectric figure of merit, ZT, and is defined as, $ZT = \frac{\sigma_o S^2}{\kappa} T$ [53]. It knows that in order to achieve a high thermoelectric material figure-of-merit, one requires a high electrical conductivity (perfect conducting medium) and a low thermal conductivity (indicator fractional order α).
- Previously, the discontinuity of the stress distribution was a critical situation and no one has explained physically the reason, while in the context of the new theory of thermoelasticity with fractional order heat transfer, the stress function is continuous.
- This paper indicates that the generalized theory of electro-thermoelasticity of fractional order heat transfer describes the behavior of an elastic body more realistically than the theory of generalized thermoelasticity with integer order.
- According to this new theory, we have to construct a new classification for materials according to their fractional parameter α where this parameter becomes a new indicator of its ability to conduct heat in conducting medium.
- The result provides a motivation to investigate conducting thermoelectric materials as a new class of applicable thermoelectric materials.

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